SSJS: Toward Transferable Adversarial Attacks in Hyperspectral Image Classification

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***Abstract*—Transferable adversarial attacks pose a significant security threat to deep neural networks (DNNs) by using a known source model to craft adversarial examples that can deceive unknown target models. However, most existing transferable attack methods are designed for natural images and face significant challenges when applied to high-dimensional data. Hyperspectral images (HSIs), characterized by high dimensionality and strong spectral redundancy, pose particular difficulties for transferable attacks. On the one hand, different HSI classification models rely on different discriminative spectral bands; thus, perturbations optimized for a source model often fail to activate the decision boundaries of a target model. On the other hand, conventional attacks generate high-frequency, pixel-level noise that lacks the spatial semantic consistency, making them susceptible to simple preprocessing defenses. To address these challenges, we propose a novel spatial–spectral joint smoothing (SSJS) attack. SSJS enhances transferability by promoting local smoothing of adversarial perturbations, which reduces the sensitivity of attacks to diverse discriminative spectral patterns. Specifically, we introduce spatial Gaussian smoothing to ensure structural consistency and multiscale weighted spectral smoothing to capture cross-band dependencies and resist wavelength shifts. Furthermore, a random masking strategy with gradient aggregation is employed to suppress source-specific noise and extract generalized vulnerabilities. Extensive experiments demonstrate that the adversarial examples generated by the proposed method achieve effective transferability. The average transferability attack success rate of the proposed method has increased by 5.46%, 6.55%, and 2.48% on three HSI datasets compared to baselines. By exposing these advanced vulnerabilities, our work provides crucial insights for developing more resilient and trustworthy deep learning models for high-dimensional data applications. The code is available at https://github.com/April-cq/HyperspectralAdversarialAttack_SSJS**

***Index Terms*—Adversarial attacks, adversarial transferability, hyperspectral image (HSI) classification.**

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I. Introduction

HYPERSPECTRAL images (HSIs) provide a wealth of spatial–spectral information [1], enabling precise material identification in fields such as meteorology [2], environmental monitoring [3], agriculture [4], and military [5]. HSI classification has evolved from classical machine learning algorithms, such as support vector machines (SVMs) and random forests (RFs), to advanced deep neural networks (DNNs) that have significantly pushed the boundaries of the field [6], [7]. However, previous research also indicates several potential security problems existing in DNN models [8], [9], [10].

Szegedy et al. [11] first discovered that adding tiny and imperceptible perturbations to a clean sample can mislead DNNs. Since then, many adversarial attack methods have been proposed to expose network flaws and improve robustness. Depending on how much model information the attacker can access, these methods are divided into two categories: white-box attacks and black-box attacks. With deeper research, the more difficult and more realistic black-box attacks have become a focus. Black-box attacks can be further divided into query-based attacks [12] which estimate gradients by repeatedly calling the interface, and transfer-based attacks [13] which generate adversarial examples on a source model and directly transfer them. Compared to query-based attacks, transfer-based attacks do not need to access the target model interface frequently and therefore are more flexible and efficient in practical applications.

However, existing transferable adversarial attacks are developed mainly on one- [14], [15] or three-band [16], [17] natural images and are not well suited for HSIs. Moreover, HSIs usually contain hundreds of narrow bands, and spectral redundancy and high correlation greatly increase gradient noise. Therefore, traditional transfer strategies show a sharply reduced success rate on black-box models [18]. Furthermore, the discriminative spectra that different HSI classification models focus on are not the same [19]. In such a "spectral-attention misalignment" situation, perturbations that mainly follow the spectral focus of the source model often fail to activate discriminative features in the target model, which causes cross-model transfer failure. Meanwhile, HSI classification models based on convolutional neural networks (CNNs) and Transformer architectures rely heavily on spatial context, untreated high-frequency pixel-level perturbations appear as discrete speckles in space, and can be easily suppressed by

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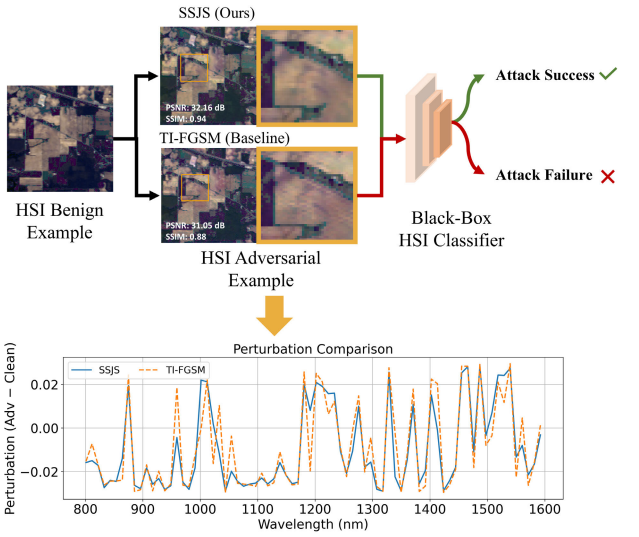


Fig. 1. Transferability comparison of adversarial examples generated by the proposed SSJS and the baseline TI-FGSM on the Indian Pines dataset. Both methods perturb a benign HSI based on the source model SSFTT. The SSJS example achieves higher PSNR and SSIM, indicating better visual fidelity, while also successfully misleading the black-box classifier. In contrast, the perturbation crafted by TI-FGSM fails to achieve a transferable attack. The spectral perturbation profiles show that SSJS yields smooth perturbations across bands, whereas TI-FGSM produces noisy fluctuations, explaining the superior transferability of SSJS.

image preprocessing technologies, such as spectral filtering and spatial resizing. Perturbations that lack spatial coherence not only have difficulty maintaining a stable gradient direction in the source model, but also cannot resist the translation and cropping differences of receptive fields in the target model.

To address the aforementioned challenges in transferable adversarial attacks on HSI classification, we propose a novel method called spatial–spectral joint smoothing (SSJS) attack, which achieves highly transferable attacks by enhancing smoothness of adversarial perturbations in spectral and spatial dimensions. Unlike traditional adversarial attack methods, SSJS first aggregates gradients computed under an ensemble of random masks at each iteration to suppress source-model-specific noise and to emphasize vulnerabilities that are common across models. The aggregated gradient is then processed by two complementary smoothing branches: a translationally invariant 2-D Gaussian convolution in the spatial domain to enforce spatial coherence, and a multiscale Gaussian smoothing across spectral bands with normalized scale weights to maintain perturbation stability under slight wavelength shifts. The fused gradients generate smoothed perturbations that alleviate overfitting to the source model and reduce sensitivity to discriminative patterns across different classification models, which enhances the transferability of HSI adversarial examples and improves robustness to preprocessing-based defenses. As illustrated in Fig. 1, compared to the state-of-the-art adversarial attack method TI-FGSM [20], the perturbations generated by our proposed SSJS approach are smoother, less perceptible to the human eye, and exhibit greater consistency and coherence. Compared with existing methods, our approach considers perturbation smoothness in both spectral and spatial dimensions, which more effectively improves attack transferability and provides stronger resistance to image preprocessing-based defenses.

The main contributions are summarized as follows.

1) We propose a novel transferable adversarial attack method for HSI classification, named SSJS. It exhibits both spectral translation invariance and spatial coherence, which improves cross-model transferability.
2) Since SSJS relies only on input-layer gradients, it can be seamlessly embedded into any existing gradient-based attack framework, promoting the effect of transfer attack.
3) We conduct evaluation experiments on three public hyperspectral datasets and six HSI classification models. SSJS increases the black-box attack success rate by 2%–7% on average compared with the best-performing baselines. Ablation experiments and visualization analysis further verify the effectiveness of each module.

The remainder of this article is organized as follows. Section II reviews related works relevant to this study. Section III provides a detailed explanation of the proposed adversarial attack methodology. Section IV describes the experiment setup and summarizes the experimental results. Section V concludes this study.

II. Preliminaries and Related Works

A. HSI Classification Based on Deep Learning

Unlike traditional machine learning methods that require handcrafted features, DNNs can directly learn spectral—spatial representations from labeled HSI data, making it a major research focus. Early studies mainly employed CNNs. For example, Makantasis et al. [21] encoded spatial and spectral information using a CNN with an MLP classifier, while Zhang et al. [22] proposed a multidimensional CNN for automatic multilevel feature extraction. However, the fixed local receptive field of convolution kernels is difficult to capture long-range correlations across pixels and bands, and deeper networks easily suffer from gradient vanishing and parameter inflation. To overcome these limitations, researchers have recently introduced the self-attention mechanism into the HSI field and built a series of Transformer-based classification models: the spatial-pooling Transformer of Ma et al. [23], the memory-enhanced Transformer of Sun et al. [24], the lightweight GSCViT of Zhao et al. [25], and the multiscale cascaded Transformer of Shi et al. [26]. These models rely on global self-attention to dynamically model long-distance spectral–spatial dependencies, and they significantly surpass CNN baselines on public datasets. In this study, we select six representative HSI classification models to evaluate the cross-model transferability of the proposed SSJS attack.

B. Adversarial Attacks

Adversarial attacks aim to expose and exploit the vulnerability of learned classifiers under deliberately crafted input perturbations. Let x be a benign example with a true label y, and $f(x;\theta)$ the classifier with parameters θ that outputs the prediction result. The goal of adversarial attack is to generate an adversarial example: $x^{adv} = x + \epsilon$ by adding

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imperceptible noise such that the model is vulnerable, i.e., $f(x^{adv};\theta) \neq y$. Generally, the noise ϵ is regularized by an ℓ_p-norm. Then, crafting an adversarial example can be formulated as an optimization problem in the following manner:

$$\arg\max_{x^{adv}} J\left(x^{adv}, y; \theta\right), \quad \text{s.t.} \left\|x^{adv} - x\right\|_p \leq \epsilon \tag{1}$$

where $J(\cdot,\cdot)$ is used as the cross-entropy loss to measure the distance between the predicted label and the true label, and p is set to ∞ in this work.

Numerous adversarial attack methods have been proposed in recent years, including gradient-based methods [14], [16], [27], [28], optimization-based methods [17], score-based methods [12], [29], and decision-based methods [30], [31]. Among them, gradient-based methods are the most representative and we provide a brief overview as follows.

1) Fast Gradient Sign Method [14]: Fast gradient sign method (FGSM) is the first gradient-based attack that crafts an adversarial example x^{adv} by maximizing the loss function $J\left(x^{adv}, y; \theta\right)$ with a one-step update

$$x^{adv} = x + \epsilon \cdot \text{sign}\left(\nabla_x J(x, y; \theta)\right) \tag{2}$$

where $\nabla_x J(x, y; \theta)$ is the gradient of loss function with respect to x and sign$(\cdot)$ denotes the sign function.

2) Iterative Fast Gradient Sign Method (I-FGSM) [28]: I-FGSM extends FGSM to an iterative version with a small step size α

$$\begin{aligned} x_{t+1}^{adv} &= x_t^{adv} + \alpha \cdot \text{sign}\left(\nabla_{x_t^{adv}} J\left(x_t^{adv}, y; \theta\right)\right) \\ x_0^{adv} &= x. \end{aligned} \tag{3}$$

3) Momentum Iterative Fast Gradient Sign Method [27]: Momentum iterative fast gradient sign method (MI-FGSM) integrates the momentum into I-FGSM and achieves much higher transferability

$$\begin{aligned} g_{t+1} &= \mu \cdot g_t + \frac{\nabla_{x_t^{adv}} J(x_t^{adv}, y; \theta)}{\|\nabla_{x_t^{adv}} J(x_t^{adv}, y; \theta)\|_1} \\ x_{t+1}^{adv} &= x_t^{adv} + \alpha \cdot \text{sign}(g_{t+1}) \end{aligned} \tag{4}$$

where $g_0 = 0$ and μ is the decay factor.

However, unlike natural images, HSIs pose additional challenges due to their long spectral dimension and spectral-spatial dependency, requiring perturbations to maintain both spectral smoothness and spatial coherence for cross-model transferability. In this work, we focus on attack transferability on HSIs.

C. Adversarial Attacks and Defense in Remote Sensing

In recent years, the study of adversarial examples has expanded from natural scenes to remote sensing (RS) images. Chen et al. [32] demonstrated the vulnerability of CNN-based RS classifiers using classical attack methods, while Li et al. [33] and Du et al. [34] investigated adversarial sensitivity in SAR image classification and recognition. Shi et al. [35] further exploited weaknesses of multimodal RS classifiers through a collaborative attack network. However, most existing methods focus on global perturbations, which are easily mitigated by image preprocessing defenses.

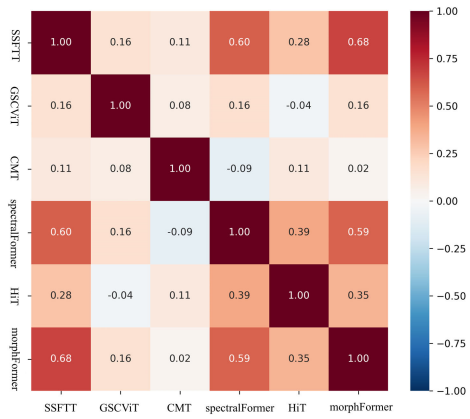


Fig. 2. Pearson correlations between spectral importance maps of different HSI classifiers.

For HSIs, research on adversarial attacks and defenses has only just begun. Liu et al. [36] proposed a local pixel-level attack to balance effectiveness and efficiency. On the defense side, Shi et al. [37] introduced an attention-based decomposition framework, and Xu et al. [38] and [39] improved robustness by exploiting global context and spatial–spectral self-attention. Nevertheless, these methods require access to the target model or internal features, which limits their applicability in black-box settings. To date, research on black-box transfer attacks for HSIs remains limited, with representative efforts by Liu et al. [18] and Shi et al. [19]. Overall, research on improving the transferability of adversarial attacks in HSI classification remains limited, and we consider this to be a worthwhile research direction.

III. Methodology

In this section, we first clarify the motivation behind the proposed approach and then detail its core components. Finally, we illustrate the attack process with an integrated spectral–spatial pipeline.

A. Motivation

Let $f_s(x;\theta_s)$ denote the source model accessible to the attacker, and $f_t(x;\theta_t)$ denote the unknown target model, where both models share the input space $\mathcal{X}$ and the output space $\mathcal{Y}$. Given an input sample $x \in \mathcal{X}$ with a ground-truth label $y \in \mathcal{Y}$, the attacker first generates an adversarial example x^{adv} on the source model, subject to the constraint $\|x^{adv} - x\|_p \leq \epsilon$. If the adversarial example simultaneously satisfies the following conditions:

$$f_s\left(x^{adv};\theta_s\right) \neq y \quad \text{and} \quad f_t\left(x^{adv};\theta_t\right) \neq y \tag{5}$$

then x^{adv} is called a transferable adversarial example, and the corresponding attack is referred to as a transfer attack.

Existing studies on transferable adversarial attacks tend to neglect two key properties of HSIs. First, different from RGB images, HSIs exhibit high-dimensional and redundant spectral characteristics, leading different classification models to emphasize different discriminative spectral bands. As shown in Fig. 2, the generally low intermodel correlations

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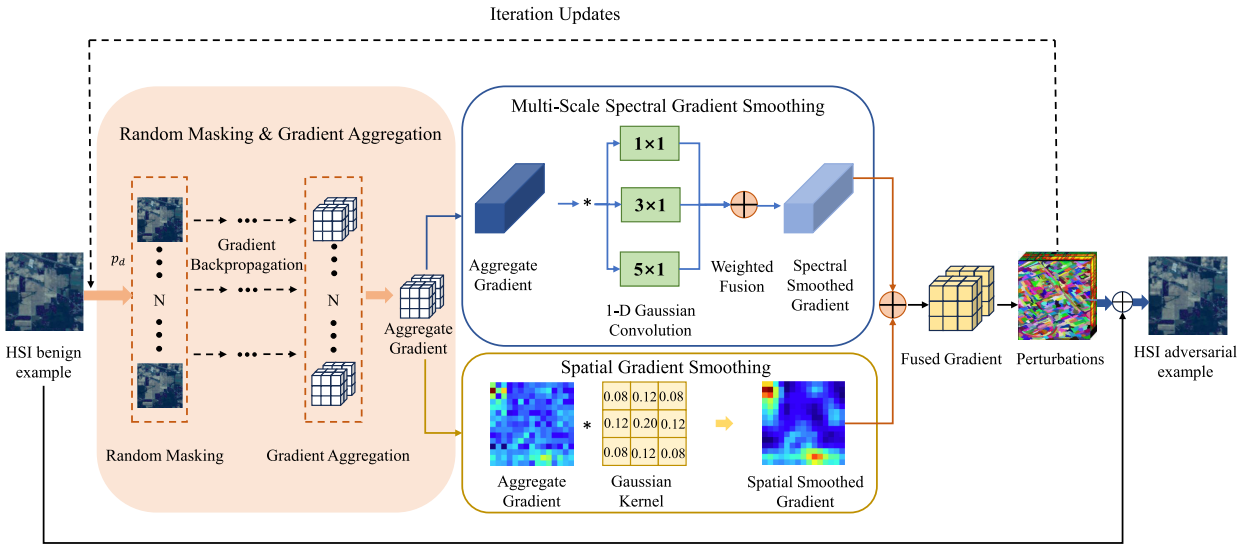


Fig. 3. Overview of the proposed hyperspectral adversarial attack framework, which incorporates random masking and gradient aggregation (RM&GA), spatial gradient smoothing (SGS), and multiscale spectral gradient smoothing (MS-SGS) to generate transferable adversarial examples.

of spectral attention distributions indicate substantial misalignment, which can hinder the transferability of adversarial perturbations. Second, modern convolutional and transformer architectures rely heavily on spatial context; adversarial perturbations that lack spatial coherence can therefore be readily suppressed by rudimentary filtering or super-pixel smoothing.

To address these challenges, We propose a joint spatial–spectral attack called SSJS, as shown in Fig. 3. The method uses gradient aggregation with random masking to reduce model-specific noise and improve generalization. It applies Gaussian smoothing in the spatial domain to enforce spatial consistency. It also applies multiscale smoothing in the spectral domain to reduce band misalignment and model dependence. By combining spatial and spectral processing, SSJS generates perturbations that are spatially consistent and spectrally adaptive, which improves transferability across different HSI classifiers.

B. Random Masking and Gradient Aggregation

To alleviate the overfitting of adversarial attacks to the source-model gradients and to improve the transferability of adversarial examples, following the idea of FIA [40], multiple randomly masked versions of the current adversarial example are constructed, and the gradients from these variants are aggregated to obtain a more robust update direction in each iteration. This operation acts as a denoising process, since model-specific noisy gradients tend to cancel out during aggregation, while stable, semantically relevant gradients are reinforced. As a result, the aggregated gradient better captures transferable feature importance across models.

More concretely, we apply a random mask to the adversarial example x^{adv} as

$$x^{mask} = M^{p_d} \odot x^{adv} \tag{6}$$

where p_d is a drop probability and M^{p_d} is an independently sampled Bernoulli mask that has the same shape as x^{adv}, and $\odot$ denotes the Hadamard product.

For the nth masked sample x_n^{mask}, we calculate its corresponding model loss $L\left(f\left(x_n^{mask}\right), y\right)$ and backpropagate to obtain the gradient. Therefore, the aggregate gradient can be expressed as

$$G^{agg} = \frac{1}{N}\sum_{n=1}^{N} \nabla_x L\left(f\left(x_n^{mask}\right), y\right) \tag{7}$$

where N denotes the number of masked samples.

The aggregated gradient G^{agg} replaces the single-sample gradient and reduces model bias. In SSJS, it is used as the initial gradient and is refined by spatial smoothing and multiscale spectral smoothing to obtain perturbations that are spatially consistent and spectrally adaptive.

C. Spatial and Spectral Smoothing

In the natural image domain, translation-invariant (TI) attacks [20] enhance the spatial smoothness and translation robustness of adversarial perturbations by convolving the gradient with a Gaussian kernel. Inspired by this idea, we design a dual-domain smoothing strategy to guarantee that the generated perturbations remain effective in both spectral and spatial dimensions.

1) Spatial Gradient Smoothing: In the spatial domain, local regions of HSIs generally exhibit structural consistency. Directly using raw gradients may cause noisy perturbations lacking spatial coherence, which could be easily mitigated by spatial resizing or denoising. To mitigate this, we convolve the aggregated gradient G^{agg} of the adversarial sample with the separable 2-D kernel K_σ to obtain the smoothed gradient

$$\tilde{G}^{spa} = K_\sigma * G^{agg} \tag{8}$$

where * denotes convolution and the separable 2-D Gaussian kernel is defined as

$$K_\sigma(i, j) = \frac{1}{2\pi\sigma^2} \cdot \exp\left(-\frac{i^2 + j^2}{2\sigma^2}\right). \tag{9}$$

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Here, σ denotes the standard deviation and $i, j \in [-((k-1)/2), ((k-1)/2)]$ denotes the kernel index offsets, where k denotes the kernel size.

By applying the 2-D Gaussian kernel along the spatial dimensions of the gradient, the gradient energy diffuses to neighboring pixels, producing a smoother adversarial perturbation.

2) Multiscale Spectral Gradient Smoothing: Unlike natural images, HSIs contain dense spectral signatures that are highly correlated across contiguous bands. To alleviate spectral-band misalignment and capture cross-band dependencies, we further apply Gaussian smoothing along the spectral dimension. Given a set of 1-D Gaussian kernels $\{K_\sigma^{(s)}\}_{s=1}^{S}$ with different receptive fields and normalized weights $\{w_s\}$ (satisfying $\sum_{s=1}^{S} w_s = 1$), the spectral-smoothed gradient is formulated as

$$\tilde{G}^{spe} = \sum_{s=1}^{S} w_s \cdot \left(K_\sigma^{(s)} * G^{agg}\right). \tag{10}$$

This multiscale formulation captures both fine-grained and broadband spectral variations, ensuring that the perturbations remain spectrally adaptive and less sensitive to model-specific preprocessing pipelines.

3) Gradient Fusion: Finally, the spatially smoothed gradient $\tilde{G}^{spa}$ and the spectrally smoothed gradient $\tilde{G}^{spe}$ are linearly fused to obtain the final update direction

$$\tilde{G} = \beta \cdot \tilde{G}^{spa} + (1-\beta) \cdot \tilde{G}^{spe} \tag{11}$$

where $\beta \in [0, 1]$ balances the contributions from spatial and spectral components. This weighted fusion leverages complementary cues: spatial smoothing preserves local structural coherence, while spectral smoothing enforces band-wise consistency, yielding perturbations that are both spatially coherent and spectrally stable.

D. Attack Algorithms

Since the proposed SSJS relies only on the input-level gradients back-propagated from the source model, it can be integrated into any gradient-based adversarial attack method to perform transferable attacks. Furthermore, SSJS is inherently architecture-agnostic. As it operates independently of model-specific intermediate features or architectural assumptions, the proposed strategy generalizes well across different HSI classifiers, including CNNs, Vision Transformer-based models, and encoder–decoder frameworks. To illustrate the workflow of transferable adversarial attack, we choose MI-FGSM [27] as the baseline optimizer and integrate it into our spatial–spectral perturbation generation framework, since its momentum mechanism stabilizes iterative updates and enhances transferability. The complete SSJS procedure as shown in Algorithm 1. We define the clean input sample as $x \in \mathbb{R}^{C\times H\times W}$ with label y, where C, H, and W represent the number of spectral bands, image height, and image width, respectively. At each iteration $t = 1, \ldots, T$, N independent Bernoulli masks $\{M_n\}_{n=1}^{N}$ are generated on the current adversarial example x_{t-1}^{adv}. Then, the corresponding gradients are computed, and the aggregate gradient G_{t-1}^{agg} is obtained according to (7), which reduces model-specific noise and strengthens semantic consistency. Next, the aggregated gradient G_{t-1}^{agg} is smoothed in both spatial and spectral domains. SGS is applied as in (8), while multiscale spectral smoothing is performed as in (10). The two smoothed gradients are combined by gradient fusion (11) to produce the final update direction $\tilde{G}_{t-1}$. This operation avoids the defect of translation variance and weakens the inhibitory effect of defensive smoothing preprocessing. Finally, the smoothed gradient $\tilde{G}_{t-1}$ is normalized by the L_1 norm and accumulated into the momentum vector. The perturbation is generated and the adversarial sample is obtained by

$$\begin{aligned} g_t &= \mu \cdot g_{t-1} + \frac{\tilde{G}_{t-1}}{\|\tilde{G}_{t-1}\|_1} \\ x_t^{adv} &= \mathrm{Clip}_{x,\epsilon}\left(x_{t-1}^{adv} + \alpha \cdot \mathrm{sign}(g_t)\right) \end{aligned} \tag{12}$$

where $\mu \in [0, 1]$ is the momentum factor, α is the step size, g_t denotes the momentum-based gradient accumulation, and $\mathrm{Clip}_{x,\epsilon}(\cdot)$ projects the perturbation into the valid ϵ-ball under the ℓ_p-norm constraint.

Algorithm 1 SSJS Attack

Input : clean HSI example x, label y; source model f; perturbation budget ϵ, iterations T; step size $\alpha = \epsilon/T$; momentum factor μ; mask number N, drop probability p_d; spatial Gaussian kernel K_σ of size $k \times k$; spectral kernel sizes $\{s\}_{s=1}^{S}$, and weights $\{w_s\}_{s=1}^{S}$ with $\sum_{s=1}^{S} w_s = 1$; fusion weight $\beta \in [0, 1]$.

Output: Adversarial example x^{adv}.

Initialize $g_0 = 0$, $x_0^{adv} = x$
for $t = 1$ **to** T **do**
  // Gradient aggregation over random masks
  $G_{t-1}^{\text{agg}} \leftarrow 0$
  for $n = 1$ **to** N **do**
    $M_n \sim \text{Bernoulli}(1 - p_d)$ (same shape as x)
    $G^n = \nabla_x \mathcal{L}(f(M_n \odot x_{t-1}^{adv}), y)$
    $G_{t-1}^{agg} \leftarrow G_{t-1}^{agg} + G_n$
  end
  $G_{t-1}^{agg} \leftarrow G_{t-1}^{agg}/N$
  // Spatial gradient smoothing
  $\widetilde{G}_{t-1}^{\text{spa}} \leftarrow K_\sigma * G_{t-1}^{agg}$
  // Multiscale spectral gradient smoothing
  for $s = 1$ **to** S **do**
    $G_s^{spe} \leftarrow K_\sigma^{(s)} * G_{t-1}^{agg}$
  end
  $\widetilde{G}_{t-1}^{spe} \leftarrow \sum_{s=1}^{S} w_s G_s^{spe}$
  // Gradient fusion
  $\widetilde{G}_{t-1} \leftarrow \beta \cdot \widetilde{G}_{t-1}^{spa} + (1-\beta) \cdot \widetilde{G}_{t-1}^{spe}$
  // momentum update
  $g_t = \mu \cdot g_{t-1} + \frac{\tilde{G}_{t-1}}{\|\tilde{G}_{t-1}\|_1}$
  // update HSI adversarial example
  $x_t^{adv} = \mathrm{Clip}_{x,\varepsilon}\left(x_{t-1}^{adv} + \alpha \cdot \mathrm{sign}(g_t)\right)$
end
return $x^{adv} = x_T^{adv}$

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Fig. 4. (a) False color images of the Indian Pines dataset. (b) Ground-truth map. (c) Labels of the Indian Pines dataset.

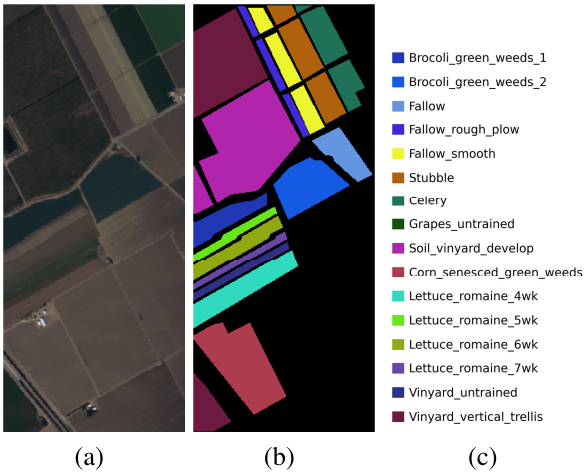


Fig. 5. (a) False color images of the Salinas dataset. (b) Ground-truth map. (c) Labels of the Salinas dataset.

IV. Experiments

In this section, we conduct experiments on three real HSI datasets and six HSI models to verify the effectiveness of our method.

A. Experiment Setup

1) Datasets: We conduct experiments on three widely used HSI datasets: Indian Pines, Salinas, and HoustonU 2018. Figs. 4–6 display the false color image, ground-truth map, and labels for each dataset, respectively.

The Indian Pines dataset was acquired by the AVIRIS sensor over the Indian Pines region in Northwest Indiana, USA, in 1992. It comprises 10 249 labeled samples that span 16 mutually exclusive land-cover classes. The spatial resolution is 20 m/pixel, and the image size is 145 × 145 pixels. The spectral range covers wavelengths from 0.4 to 2.5 μm over 224 bands. After removing 24 noisy and water-absorption bands, 200 bands remain for classification.

The Salinas dataset was collected by the AVIRIS sensor in the Salinas Valley, California, USA. It includes 54 129 labeled samples classified into 16 land-cover classes. The image size is 512 × 217 pixels at a spatial resolution of 3.7 m/pixel, with data in 224 spectral bands. After filtering out noisy bands, 204 bands are reserved for experiments.

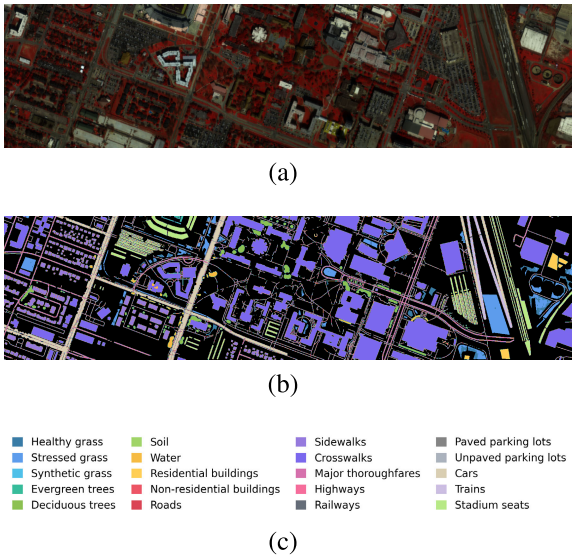


Fig. 6. (a) False color images of the HoustonU 2018 dataset. (b) Ground-truth map. (c) Labels of the HoustonU 2018 dataset.

TABLE I
Training and Testing Sample Counts for Indian Pines, Salinas, and HoustonU 2018 Datasets

Indian Pines			Salinas			HoustonU 2018		
Class	Train	Test	Class	Train	Test	Class	Train	Test
1	9	32	1	402	1406	1	1960	6859
2	285	1000	2	745	2608	2	6501	22751
3	166	581	3	395	1383	3	137	479
4	47	166	4	279	976	4	2719	9517
5	97	338	5	535	1875	5	1004	3515
6	146	511	6	792	2771	6	903	3161
7	5	20	7	716	2505	7	53	186
8	95	335	8	2254	7890	8	7955	27840
9	4	14	9	1241	4342	9	44750	156627
10	195	680	10	655	2295	10	9173	32106
11	491	1719	11	214	748	11	6806	23820
12	119	415	12	385	1349	12	303	1063
13	41	143	13	184	641	13	9269	32444
14	253	886	14	214	749	14	1973	6906
15	77	270	15	1453	5088	15	1387	4856
16	19	65	16	361	1265	16	2300	8050
						17	29	102
						18	1309	4583
						19	1074	3758
						20	1365	4777

The HoustonU 2018 dataset was acquired by the HSI CASI 1500 over the region of the University of Houston on February 16, 2017. The spatial resolution is 1 m and the spatial size of HoustonU 2018 is 601 × 2384 with 48 spectral bands. The dataset contains 20 available classes of interest.

2) Training Examples: To form each sample, we extract a 15 × 15 spatial neighborhood around every pixel as input. The exact numbers of training and test samples for each dataset are listed in Table I. Our experimental setup follows the standard evaluation protocol in the HSI adversarial attack field, to ensure a fair comparison with state-of-the-art methods.

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3) Networks: We train six popular classification networks proposed in recent years: SSFTT[1] [41], GSCViT[2] [25], CMT[3] [44], SpectralFormer[4] [45], morphFormer[5] [42], and HiT[6] [43] to assess the transferability of our adversarial attacks. These models span from hybrid CNN-Transformer structures to pure Transformer-based encoders. SSFTT [41] is a typical hybrid model that combines CNN and transformer for spectral–spatial and semantic features. GSCViT [25] employs groupwise separable convolutions for efficient global–local feature capture. CMT [44] uses multiscale convolutional embedding and masked pretraining for contextual representation. SpectralFormer [45] emphasizes spectral embedding within a transformer architecture. MorphFormer [42] integrates morphological convolutions and transformers for structural feature extraction. HiT [43] leverages spectral-adaptive 3-D convolutions and permutation to capture subtle spectral and spatial contexts.

4) Baseline Attacks: To demonstrate the effectiveness of the proposed SSJS, we compare it with seven well-known attack methods: FGSM [14], MI-FGSM [27], SS-FGSM [19], VMI-FGSM [46], DI-FGSM [47], TI-FGSM [20], and the spectrally extended TI-FGSM (S-TI-FGSM). Specifically, FGSM [14] perturbs inputs using the gradient sign of the loss function. MI-FGSM [27] incorporates momentum into FGSM to stabilize perturbation direction. SS-FGSM [19] promotes smoother, locally consistent perturbations by enforcing spatial and spectral constraints to enhance transferability and robustness. VMI-FGSM [46] enhances MI-FGSM by adjusting variance to strengthen perturbations. DI-FGSM [47] uses random input transformations to diversify adversarial examples, improving transferability. TI-FGSM [20] adds translation invariance to perturbations, increasing the robustness across spatial positions. S-TI-FGSM further extends TI-FGSM by additionally applying a 1-D Gaussian smoothing kernel along the spectral dimension.

In addition, we apply three well-known image-processing-based defense methods to verify attack robustness, including spatial resizing [48], spectral smoothing filtering [49], and principal component analysis (PCA)-based dimensionality reduction, to evaluate robustness against both spatial- and spectral-domain preprocessing.

5) Parameters: In all experiments, the perturbation budget ϵ is set to 0.03, the number of iterations T is 10, the momentum μ is set to 1.0, and the step size is $\alpha = \epsilon/T$. For the baseline attacks, the transform probability in DI-FGSM is 0.5, the kernel size in TI-FGSM and S-TI-FGSM is 3, and we use superpixels of size 4 and spectral clusters of size 20 in SS-FGSM. In our SSJS, the drop probability p_d is 0.01, the mask number N is 10 for gradient aggregation, the spatial Gaussian kernel size k is 3, the spectral multiscale Gaussian kernels with sizes {1,3,5} and the corresponding weights are set to [0.2, 0.3, 0.5]. And the spatial and spectral gradients are fused with a weighting factor $\beta = 0.3$.

[1]https://github.com/zgr6010/HSI_SSFTT
[2]https://github.com/flyzzie/TGRS-GSC-VIT
[3]https://github.com/rookie-YIFAN/CMT
[4]https://github.com/danfenghong/IEEE_TGRS_SpectralFormer
[5]https://github.com/mhaut/morphFormer
[6]https://github.com/xiachangxue/DeepHyperX

B. Attack Transferability Comparison

To validate the effectiveness and transferability of the proposed SSJS method, we conduct comparative experiments against seven state-of-the-art baseline methods across six models and three datasets. In line with standard adversarial attack benchmarks, our experiments focus on cross-model, intradataset transferability. To quantitatively assess attack transferability, we adopt the attack success rate as the evaluation metric, which is defined as the percentage of adversarial samples that successfully mislead the target model. To ensure the reliability of the comparative results, we conduct each experiment over five independent runs with different random seeds. The mean values and standard deviations of the attack success rates are summarized in Tables II–IV, respectively. Each table lists the source models in the first column and the corresponding target models in the first row.

1) Indian Pines Dataset: Table II shows the attack success rates on the Indian Pines dataset. In general, SSJS consistently achieves superior success rates in both white-box and black-box scenarios. Specifically, for the white-box setting, SSJS attains nearly perfect attack rates, outperforming other baselines including MI-FGSM, DI-FGSM, and TI-FGSM. This demonstrates the strong optimization capacity and attack strength of our method in directly perturbing the source model. In the more challenging black-box setting, SSJS significantly outperforms the baselines in transfer attack performance. For instance, compared with the enhanced baseline S-TI-FGSM, SSJS consistently demonstrates superior performance in transfer attacks. Overall, SSJS achieves an average improvement of 5.46% in transfer attack success rate over the best-performing baseline methods. These results confirm that SSJS is not only powerful in the white-box setting but also exhibits remarkable generalization to unseen black models, making it a promising threat model for real-world HSI classification systems.

2) Salinas Dataset: The experimental results on the Salinas dataset are shown in Table III. Unlike Indian Pines, Salinas offers higher spatial resolution and more structured, contiguous regions of homogeneous crops. This provides a more challenging scenario for transfer-based attacks. Nevertheless, our methods still demonstrate an effective improvement in the attack transferability of the baseline attacks in most cases. In the white-box setting, SSJS attains the highest success rate when morphFormer is used as the source model, demonstrating its optimization capability on complex architectures. In the black-box setting, SSJS consistently performs better than baseline methods, including S-TI-FGSM. On average, SSJS improves the transfer attack success rate by 6.55% compared with the best-performing baselines, indicating strong transferability and stability. This improvement is attributed to the joint spatial–spectral design of SSJS, which enables perturbations to generalize across different model architectures.

3) HoustonU 2018 Dataset: As shown in Table IV, SSJS consistently yields among the highest success rates across nearly all black-box transfer scenarios, demonstrating superior transferability. It is worth noting that on the HoustonU

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TABLE II

SUCCESS RATE (%) OF DIFFERENT ATTACKS AGAINST SIX TRAINED NETWORKS ON THE INDIAN PINES DATASET. "*" INDICATES A WHITE-BOX ATTACK SCENARIO, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	Attack	SSFTT	GSCViT	CMT	spectralFormer	HiT	morphFormer
SSFTT [41], 2022	MI-FGSM	98.02 ± 0.09*	46.59 ± 0.13	94.20 ± 0.10	49.35 ± 0.17	73.35 ± 0.31	70.55 ± 0.30
	SS-FGSM	**100.00 ± 0.00***	34.98 ± 0.50	93.62 ± 0.21	31.83 ± 0.36	59.57 ± 0.23	48.86 ± 0.31
	VMI-FGSM	98.99 ± 0.07*	49.50 ± 0.10	94.31 ± 0.08	51.22 ± 0.22	76.02 ± 0.26	70.62 ± 0.30
	FGSM	95.65 ± 0.08*	42.73 ± 0.21	91.40 ± 0.16	43.71 ± 0.26	61.85 ± 0.30	57.03 ± 0.27
	DI-FGSM	99.31 ± 0.04*	33.66 ± 0.22	93.66 ± 0.05	36.10 ± 0.32	57.77 ± 0.49	57.07 ± 0.38
	TI-FGSM	98.81 ± 0.04*	50.03 ± 0.30	94.15 ± 0.10	51.88 ± 0.15	76.92 ± 0.22	73.92 ± 0.22
	S-TI-FGSM	98.10 ± 0.02*	51.61 ± 0.12	95.24 ± 0.09	57.94 ± 0.05	83.57 ± 0.43	74.31 ± 0.34
	SSJS	**100.00 ± 0.00***	**58.86 ± 0.33**	**96.45 ± 0.12**	**58.39 ± 0.37**	**88.51 ± 0.13**	**76.73 ± 0.11**
GSCViT [25], 2024	MI-FGSM	18.32 ± 0.25	87.36 ± 0.15*	36.48 ± 0.25	20.02 ± 0.46	19.01 ± 0.21	12.77 ± 0.23
	SS-FGSM	27.93 ± 0.34	86.87 ± 0.30*	44.10 ± 0.18	26.74 ± 0.56	28.22 ± 0.18	20.76 ± 0.58
	VMI-FGSM	19.88 ± 0.22	88.38 ± 0.18*	38.50 ± 0.20	21.91 ± 0.37	21.66 ± 0.16	15.72 ± 0.29
	FGSM	27.21 ± 0.24	78.36 ± 0.22*	37.82 ± 0.20	28.50 ± 0.59	26.65 ± 0.25	27.17 ± 0.22
	DI-FGSM	3.60 ± 0.20	92.56 ± 0.20*	14.57 ± 0.13	12.28 ± 0.50	3.97 ± 0.27	2.88 ± 0.12
	TI-FGSM	21.21 ± 0.38	92.30 ± 0.47*	36.78 ± 0.28	21.08 ± 0.24	21.77 ± 0.33	15.89 ± 0.24
	S-TI-FGSM	20.64 ± 0.09	88.21 ± 0.06*	42.65 ± 0.01	24.15 ± 0.58	24.17 ± 0.11	16.08 ± 0.26
	SSJS	**37.44 ± 0.30**	**94.87 ± 0.04***	**50.07 ± 0.43**	**37.61 ± 0.40**	**41.73 ± 0.29**	**32.85 ± 0.22**
morphFormer [42], 2023	MI-FGSM	61.08 ± 0.43	46.84 ± 0.19	84.67 ± 0.20	41.34 ± 0.43	51.31 ± 0.17	94.60 ± 0.04*
	SS-FGSM	64.57 ± 0.21	50.29 ± 0.33	86.49 ± 0.33	44.87 ± 0.33	57.88 ± 0.44	95.67 ± 0.08*
	VMI-FGSM	63.78 ± 0.41	48.76 ± 0.19	85.29 ± 0.28	43.68 ± 0.47	54.13 ± 0.32	95.10 ± 0.08*
	FGSM	50.12 ± 0.06	40.55 ± 0.32	74.57 ± 0.13	37.87 ± 0.34	43.72 ± 0.32	76.19 ± 0.15*
	DI-FGSM	41.17 ± 0.70	21.00 ± 0.45	79.50 ± 0.57	26.32 ± 0.66	27.45 ± 0.15	96.57 ± 0.13*
	TI-FGSM	69.50 ± 0.48	53.11 ± 0.44	**89.50 ± 0.40**	48.71 ± 0.59	62.59 ± 0.68	95.69 ± 0.24*
	S-TI-FGSM	64.69 ± 0.03	53.09 ± 0.17	85.94 ± 0.11	47.58 ± 0.39	60.36 ± 0.03	94.34 ± 0.05*
	SSJS	**73.41 ± 0.39**	**64.52 ± 0.49**	87.32 ± 0.06	**55.98 ± 0.18**	**73.42 ± 0.30**	**97.48 ± 0.10***
HiT [43], 2022	MI-FGSM	51.10 ± 0.33	40.11 ± 0.15	83.02 ± 0.15	63.77 ± 0.14	98.66 ± 0.06*	40.62 ± 0.29
	SS-FGSM	49.16 ± 0.43	33.43 ± 0.29	81.80 ± 0.20	57.63 ± 0.15	98.85 ± 0.14*	28.26 ± 0.41
	VMI-FGSM	51.26 ± 0.31	40.31 ± 0.15	83.27 ± 0.11	63.85 ± 0.19	98.66 ± 0.05*	40.81 ± 0.35
	FGSM	49.96 ± 0.34	42.28 ± 0.22	78.54 ± 0.19	61.19 ± 0.24	95.05 ± 0.11*	38.93 ± 0.22
	DI-FGSM	40.43 ± 0.17	25.00 ± 0.32	80.15 ± 0.23	54.12 ± 0.38	**99.23 ± 0.11***	27.52 ± 0.32
	TI-FGSM	62.92 ± 0.24	52.87 ± 0.54	90.32 ± 0.25	70.90 ± 0.13	98.80 ± 0.09*	53.31 ± 0.39
	S-TI-FGSM	64.90 ± 0.31	60.57 ± 0.06	**90.81 ± 0.02**	72.38 ± 0.04	97.70 ± 0.03*	58.70 ± 0.14
	SSJS	**68.94 ± 0.41**	**64.48 ± 0.43**	90.42 ± 0.05	**79.32 ± 0.33**	98.68 ± 0.07*	**62.10 ± 0.21**

2018 dataset, SSJS achieves an average transfer success rate improvement of only 2.48%, which is smaller than that on Indian Pines (5.46%) and Salinas (6.55%). This difference is mainly due to the intrinsic spatial–spectral characteristics of the datasets. Indian Pines and Salinas contain large homogeneous regions and lower spatial resolution, which fit well with the spatial smoothing prior of SSJS. In contrast, HoustonU 2018 has much higher spatial resolution and many man-made structures with sharp edges and rich high-frequency details. As a result, spatial smoothing is more effective for vegetation scenes than for urban scenes with complex boundaries. In addition, the hyperparameter analysis in Section IV-E shows that the optimal parameters follow similar trends across all datasets, indicating that SSJS generalizes well and is not tuned only for vegetation data.

Furthermore, we analyze the computational complexity of the compared attack methods, as shown in Table V. It reports the average time required to generate adversarial samples for one test batch (batch size is 32). Due to the integration of RM&GA, SGS, and MS-SGS modules, SSJS is more computationally intensive and consumes more execution time than the compared baseline methods. However, we consider this increased complexity a necessary and worthwhile tradeoff for achieving superior attack transferability in the challenging HSI domain.

C. Robustness Evaluation Against Defense Methods

To verify the robustness of the proposed method, we conducted experiments on three HSI datasets using three defense methods: spatial resizing [48], spectral filtering [49], and PCA-based dimensionality reduction. These preprocessing operations are commonly used to suppress adversarial perturbations and to test the robustness of transfer attacks.

1) Spatial Resizing: Spatial resizing mitigates local adversarial perturbations by means of bilinear interpolation. Considering an adversarial HSI sample of dimensions $C \times H \times W$, the defense first applies a spatial downsampling operation with resize factor κ, and then restores the image to its original resolution via spatial upsampling. Fig. 7 illustrates the attack success rates of the proposed SSJS and seven representative baselines with different resize factors ranging from 1.0 to 3.0. In this experiment, we used GSCViT as an example to assess the robustness of different attack methods because of its ability to jointly model local spectral–spatial features and global dependencies with reduced parameter complexity, while achieving state-of-the-art classification performance. The results demonstrate that larger-size spatial resizing leads to lower attack success rates as it removes more perturbations. Most baseline methods suffer a decrease of about 2%–9% when κ increases from 1.0 to 3.0. In contrast, SSJS remains largely stable over this range, with only about a 2% decrease,

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TABLE III

SUCCESS RATE (%) OF DIFFERENT ATTACKS AGAINST SIX TRAINED NETWORKS ON THE SALINAS DATASET. "*" INDICATES A WHITE-BOX ATTACK SCENARIO, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	Attack	SSFTT	GSCViT	CMT	spectralFormer	HiT	morphFormer
SSFTT [41], 2022	MI-FGSM	75.05 ± 0.07*	37.14 ± 0.07	59.12 ± 0.09	53.71 ± 0.09	34.72 ± 0.15	43.70 ± 0.04
	SS-FGSM	**98.35 ± 0.04***	43.51 ± 0.12	66.55 ± 0.20	45.34 ± 0.08	27.65 ± 0.14	40.63 ± 0.14
	VMI-FGSM	95.70 ± 0.04*	50.14 ± 0.06	76.51 ± 0.11	50.87 ± 0.09	42.80 ± 0.09	61.42 ± 0.09
	FGSM	60.20 ± 0.10*	32.82 ± 0.07	52.03 ± 0.05	40.34 ± 0.03	25.67 ± 0.04	39.81 ± 0.06
	DI-FGSM	79.31 ± 0.15*	29.46 ± 0.07	57.66 ± 0.21	43.92 ± 0.11	19.15 ± 0.11	29.46 ± 0.10
	TI-FGSM	78.07 ± 0.15*	41.61 ± 0.12	60.75 ± 0.18	54.99 ± 0.12	40.99 ± 0.13	48.69 ± 0.15
	S-TI-FGSM	74.18 ± 1.08*	50.98 ± 1.12	65.94 ± 1.36	58.69 ± 0.91	46.44 ± 1.89	55.66 ± 1.83
	SSJS	96.03 ± 0.07*	**60.48 ± 0.13**	**82.93 ± 0.16**	**62.11 ± 0.17**	**53.97 ± 0.16**	**70.63 ± 0.07**
GSCViT [25], 2024	MI-FGSM	47.12 ± 0.08	95.41 ± 0.05*	39.40 ± 0.14	51.95 ± 0.16	37.40 ± 0.07	33.08 ± 0.10
	SS-FGSM	57.70 ± 0.07	**98.02 ± 0.02***	48.33 ± 0.15	55.26 ± 0.18	50.72 ± 0.11	42.59 ± 0.06
	VMI-FGSM	48.15 ± 0.09	96.35 ± 0.03*	42.77 ± 0.08	52.81 ± 0.06	40.98 ± 0.11	37.15 ± 0.10
	FGSM	48.44 ± 0.18	83.63 ± 0.10*	37.24 ± 0.09	48.11 ± 0.11	34.94 ± 0.11	31.42 ± 0.18
	DI-FGSM	30.00 ± 0.16	96.46 ± 0.11*	26.83 ± 0.08	31.97 ± 0.07	23.84 ± 0.11	19.98 ± 0.06
	TI-FGSM	50.59 ± 0.17	90.39 ± 0.14*	42.45 ± 0.19	55.37 ± 0.13	42.84 ± 0.18	36.98 ± 0.20
	S-TI-FGSM	48.64 ± 0.38	94.17 ± 0.09*	40.24 ± 1.52	55.68 ± 1.04	38.82 ± 1.81	33.11 ± 0.69
	SSJS	**61.12 ± 0.07**	95.56 ± 0.02*	**64.52 ± 0.32**	**61.85 ± 0.17**	**58.96 ± 0.09**	**53.99 ± 0.07**
morphFormer [42], 2023	MI-FGSM	52.41 ± 0.11	35.21 ± 0.14	57.75 ± 0.10	45.21 ± 0.13	42.98 ± 0.14	85.26 ± 0.06*
	SS-FGSM	55.99 ± 0.30	41.16 ± 0.21	59.64 ± 0.24	48.75 ± 0.23	43.04 ± 0.27	88.43 ± 0.13*
	VMI-FGSM	55.76 ± 0.12	39.68 ± 0.08	61.45 ± 0.08	48.26 ± 0.13	47.08 ± 0.12	87.41 ± 0.09*
	FGSM	44.77 ± 0.09	25.31 ± 0.07	45.78 ± 0.07	33.34 ± 0.05	31.02 ± 0.07	63.30 ± 0.07*
	DI-FGSM	44.00 ± 0.17	28.36 ± 0.19	44.92 ± 0.07	31.27 ± 0.18	23.22 ± 0.19	86.58 ± 0.06*
	TI-FGSM	50.96 ± 0.12	36.19 ± 0.13	56.90 ± 0.13	40.87 ± 0.13	38.91 ± 0.12	82.08 ± 0.10*
	S-TI-FGSM	52.91 ± 2.13	44.60 ± 1.62	60.29 ± 0.80	50.90 ± 1.08	51.99 ± 1.00	86.80 ± 0.89*
	SSJS	**57.41 ± 0.12**	**50.70 ± 0.09**	**63.03 ± 0.08**	**64.48 ± 0.12**	**61.12 ± 0.07**	**89.30 ± 0.08***
HiT [43], 2022	MI-FGSM	20.09 ± 0.12	46.94 ± 0.11	25.44 ± 0.09	41.26 ± 0.08	93.60 ± 0.03*	24.00 ± 0.18
	SS-FGSM	24.56 ± 0.09	32.20 ± 0.16	22.11 ± 0.14	45.78 ± 0.08	93.50 ± 0.03*	24.73 ± 0.12
	VMI-FGSM	20.43 ± 0.17	47.20 ± 0.11	25.66 ± 0.09	41.46 ± 0.13	**93.61 ± 0.03***	41.58 ± 0.26
	FGSM	41.31 ± 0.06	53.05 ± 0.22	31.07 ± 0.17	47.97 ± 0.23	86.54 ± 0.06*	39.08 ± 0.11
	DI-FGSM	33.48 ± 0.22	54.95 ± 0.15	45.14 ± 0.22	51.56 ± 0.14	92.53 ± 0.02*	35.24 ± 0.16
	TI-FGSM	49.56 ± 0.23	67.48 ± 0.08	60.62 ± 0.09	66.61 ± 0.15	91.37 ± 0.08*	51.60 ± 0.14
	S-TI-FGSM	51.45 ± 1.79	64.95 ± 1.94	62.54 ± 0.57	**68.79 ± 1.90**	86.98 ± 1.31*	52.43 ± 0.61
	SSJS	**55.59 ± 0.08**	**68.29 ± 0.07**	**65.03 ± 0.07**	66.64 ± 0.04	93.20 ± 0.05*	**54.82 ± 0.14**

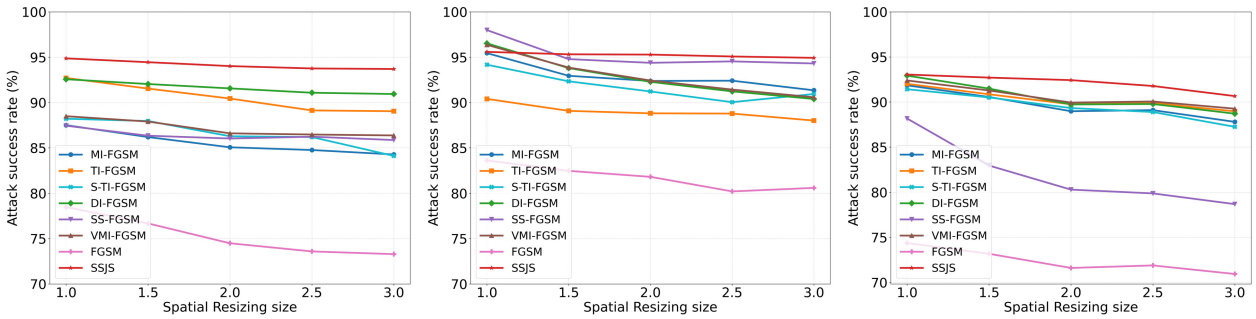


Fig. 7. Attack success rates of HSI adversarial examples with spatial resizing. The HSI adversarial examples are generated on GSCViT, and the maximum perturbation budget is 0.03. (a) Indian Pines. (b) Salinas. (c) HoustonU 2018.

which is the smallest relative degradation among all methods. This stability stems from the joint spatial–spectral smoothing in SSJS, which distributes perturbations more coherently and makes them less sensitive to moderate rescaling.

2) Spectral Filtering: The HSI has redundant spectral bands; therefore, local spectral filtering is also a challenging defense method. We apply mean filtering along the spectral dimension to verify the robustness of the attack methods. Fig. 8 shows results for SSJS and seven baselines when the filter size increases from 1 (no filtering) to 11. As the filter size increases, the attack success rates of all methods decrease. However, SSJS still maintains the highest success rates among all methods, showing the strongest robustness. This robustness comes from multiscale spectral smoothing and gradient aggregation in SSJS, which spread perturbations across neighboring bands and reduce reliance on narrow-band high-frequency noise, making them less sensitive to spectral filtering.

3) PCA-Based Dimensionality Reduction: PCA-based dimensionality reduction is a common preprocessing step in HSI classification to reduce spectral redundancy. We apply PCA to adversarial examples before feeding them into the

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TABLE IV

SUCCESS RATE (%) OF DIFFERENT ATTACKS AGAINST SIX TRAINED NETWORKS ON THE HOUSTONU 2018 DATASET. "*" INDICATES A WHITE-BOX ATTACK SCENARIO, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	Attack	SSFTT	GSCViT	CMT	spectralFormer	HiT	morphFormer
SSFTT, 2022	MI-FGSM	97.32±0.11*	64.58±0.60	84.68±0.44	48.58±0.51	56.29±0.36	74.93±0.91
	SS-FGSM	96.05±0.02*	64.30±0.55	84.43±0.27	43.83±0.53	52.20±0.65	71.44±0.79
	VMI-FGSM	97.79±0.10*	67.54±0.73	85.67±0.26	49.18±0.51	57.63±0.55	77.04±0.97
	FGSM	91.44±0.26*	59.17±0.97	77.99±0.58	46.46±0.99	53.84±0.98	68.77±1.98
	DI-FGSM	**98.24±0.13***	43.08±0.75	80.55±0.70	28.55±0.48	35.91±0.43	61.66±1.20
	TI-FGSM	97.43±0.04*	63.98±0.52	84.84±0.36	48.09±0.35	57.00±0.61	74.38±0.84
	S-TI-FGSM	97.41±0.02*	65.36±0.52	85.29±0.34	50.76±0.36	57.57±0.34	75.46±0.99
	SSJS	97.51±0.09*	**71.29±0.23**	**86.79±0.12**	**56.37±0.48**	**61.67±0.33**	**81.48±0.11**
GSCViT, 2024	MI-FGSM	70.46±0.55	91.79±0.24*	67.79±0.77	59.43±0.75	63.70±0.16	66.68±0.92
	SS-FGSM	64.18±0.44	87.99±0.18*	64.16±0.66	55.43±0.69	61.34±0.46	62.78±0.68
	VMI-FGSM	72.68±0.57	92.28±0.12*	69.83±0.92	63.66±0.57	66.81±0.41	**69.37±0.55**
	FGSM	67.73±0.88	73.93±0.53*	64.40±0.95	60.60±1.09	61.35±0.30	63.42±1.41
	DI-FGSM	61.40±0.41	92.82±0.27*	61.60±0.66	45.12±0.79	54.06±0.09	61.16±1.03
	TI-FGSM	72.90±0.32	91.68±0.24*	70.22±0.71	63.35±0.65	65.13±0.15	66.92±0.86
	S-TI-FGSM	71.57±0.53	91.42±0.16*	68.67±0.83	61.54±0.92	65.38±0.12	66.83±1.04
	SSJS	**74.88±0.12**	**93.27±0.08***	**71.91±0.04**	**66.93±0.54**	**69.61±0.27**	69.32±0.26
morphFormer, 2023	MI-FGSM	78.69±0.72	61.35±0.33	76.73±1.02	51.99±0.71	67.61±0.62	95.04±0.56*
	SS-FGSM	65.42±0.49	50.06±0.04	63.27±0.65	40.08±0.05	53.74±0.01	69.45±1.08*
	VMI-FGSM	80.52±0.77	63.85±1.12	78.31±0.72	54.48±0.61	69.26±1.29	95.07±0.96*
	FGSM	47.86±0.43	35.42±0.57	47.21±0.28	26.67±0.62	43.47±0.49	61.78±0.83*
	DI-FGSM	67.25±0.54	42.32±0.15	65.58±1.02	29.73±0.56	46.52±0.76	94.78±0.55*
	TI-FGSM	78.81±0.69	62.70±0.38	78.19±1.08	54.09±0.59	69.18±0.77	93.05±0.70*
	S-TI-FGSM	78.70±0.71	61.99±0.48	77.00±1.05	53.96±0.68	68.23±0.90	94.71±0.65*
	SSJS	**81.87±0.42**	**65.25±0.08**	**80.35±0.17**	**56.72±0.36**	**71.20±0.49**	**96.37±0.18***
HiT, 2022	MI-FGSM	65.11±0.43	55.87±0.23	68.24±0.74	45.39±0.52	94.58±0.15*	63.52±0.53
	SS-FGSM	62.13±0.50	52.77±0.14	65.40±0.61	45.62±0.73	91.68±0.18*	61.28±0.80
	VMI-FGSM	66.00±0.50	56.98±0.23	68.23±0.47	46.74±0.53	94.62±0.15*	64.03±0.52
	FGSM	60.19±0.64	54.41±0.27	61.47±0.61	49.69±0.95	83.01±0.27*	57.91±0.61
	DI-FGSM	52.68±0.61	45.13±0.49	60.15±1.09	33.98±0.37	94.41±0.15*	61.18±0.79
	TI-FGSM	67.21±0.40	61.97±0.01	71.05±0.44	52.76±0.63	92.99±0.07*	68.02±0.46
	S-TI-FGSM	68.89±0.56	58.09±0.06	70.85±0.94	49.75±1.13	94.50±0.07*	64.78±0.46
	SSJS	**70.55±0.77**	**62.66±0.06**	**72.47±0.93**	**57.23±0.88**	**94.66±0.19***	**71.30±0.52**

TABLE V

COMPUTATIONAL COMPLEXITY ANALYSIS ON THREE DATASETS [RUNNING TIME (S)]

Dataset	Model*	Attack Methods							
		FGSM	MI-FGSM	SS-FGSM	VMI-FGSM	DI-FGSM	TI-FGSM	S-TI-FGSM	SSJS
Indian Pines	SSFTTNet	0.02	0.15	1.04	0.81	0.14	0.23	0.24	1.67
	GSCViT	0.05	0.44	1.74	2.62	0.44	0.49	0.53	4.76
	morphFormer	0.05	0.37	1.51	2.04	0.35	0.43	0.45	4.66
	HiT	0.12	0.83	2.82	5.79	1.05	1.04	1.13	12.27
Salinas	SSFTTNet	0.02	0.15	1.20	0.76	0.12	0.20	0.23	1.65
	GSCViT	0.02	0.31	1.26	1.53	0.24	0.27	0.28	2.74
	morphFormer	0.03	0.40	1.37	1.85	0.30	0.36	0.37	4.74
	HiT	0.10	0.74	2.26	4.85	0.92	0.80	0.96	12.03
HoustonU 2018	SSFTTNet	0.01	0.13	0.96	0.82	0.14	0.21	0.24	1.61
	GSCViT	0.03	0.32	1.48	1.53	0.28	0.29	0.29	2.62
	morphFormer	0.04	0.42	1.32	2.11	0.31	0.38	0.39	3.30
	HiT	0.09	0.84	2.22	4.81	0.83	1.11	1.18	10.46

* Parameters (M): SSFTTNet: 0.93, GSCViT: 0.56, morphFormer: 0.20, HiT: 51.26
Test batch size: 32 GPU: NVIDIA TITAN RTX

models. The spectral dimension is reduced from 200 to 50 for Indian Pines, from 204 to 50 for Salinas, and from 48 to 20 for HoustonU 2018. The results are reported in Table VI. When PCA is applied, the attack success rates of all methods decrease, which indicates that dimensionality reduction removes part of the perturbations. However, compared with baseline attacks, SSJS shows smaller performance degradation and achieves the highest post-PCA success rates on all three datasets. On Indian Pines, Salinas, and HoustonU 2018, SSJS decreases by only about 2%, 12%, and 22%, respectively, while most baselines suffer much larger drops. Under PCA-based reduction, SSJS consistently achieves the highest attack success rates across all three datasets. This robustness is mainly due to the multiscale spectral smoothing in SSJS, which spreads perturbations across adjacent bands and preserves low-frequency spectral structures. Since PCA

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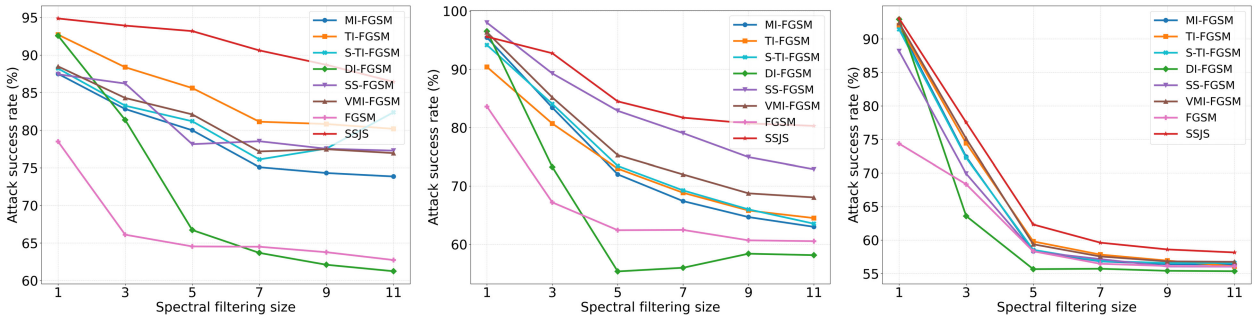


Fig. 8. Attack success rates of HSI adversarial examples with spectral filtering. The HSI adversarial examples are generated on GSCViT, and the maximum perturbation budget is 0.03. (a) Indian Pines. (b) Salinas. (c) HoustonU 2018.

TABLE VI
ATTACK SUCCESS RATE (%) UNDER PCA-BASED DIMENSIONALITY REDUCTION DEFENSE ON THREE HSI DATASETS

Attack Method	Indian Pines		Salinas		HoustonU 2018	
	Original	PCA-Reduced	Original	PCA-Reduced	Original	PCA-Reduced
MI-FGSM	87.36	79.16	95.41	72.59	91.79	61.58
SS-FGSM	86.87	79.11	**98.02**	81.66	87.99	59.27
VMI-FGSM	88.38	81.26	96.35	75.76	92.28	63.59
FGSM	78.36	66.10	83.63	66.18	73.93	60.18
DI-FGSM	92.56	70.09	96.46	66.35	92.82	55.64
TI-FGSM	92.30	84.05	90.39	71.14	91.68	62.92
S-TI-FGSM	88.21	79.76	94.17	73.36	91.42	62.24
SSJS	**94.87**	**92.42**	95.56	**83.75**	**93.27**	**71.64**

keeps principal components that capture dominant spectral patterns, such coherent perturbations are more likely to survive dimensionality reduction, while band-specific high-frequency noise produced by baselines is largely suppressed. Therefore, SSJS exhibits stronger robustness under PCA-based defenses.

D. Ablation Experiment

1) Influence of Components: This section studies the effect of each component on attack performance using ablation experiments. Due to the high computational cost, we use GSCViT as the source model and treat the other five models as target models. The parameter settings follow Section V-A5.

The experimental results are summarized in Table VII. It can be observed that enabling all three modules simultaneously yields the highest attack success rate across nearly all target models. On the Indian Pines dataset, the combination of MS-SGS, RM&GA, and SGS significantly outperforms other configurations, achieving a notable improvement. A similar trend is evident in the Salinas and HoustonU 2018 datasets, where the combined configuration generally surpasses other variations. On the Salinas dataset, the white-box success rate decreases slightly because spectral smoothing spread the perturbations and weaken fine-grained patterns learned by the source model. However, the proposed approach delivers notably stronger cross-model transferability. These findings demonstrate that each module plays a complementary role, enhancing attack transferability, and the joint usage maximizes the attack success rate, validating the effectiveness of the integrated design strategy.

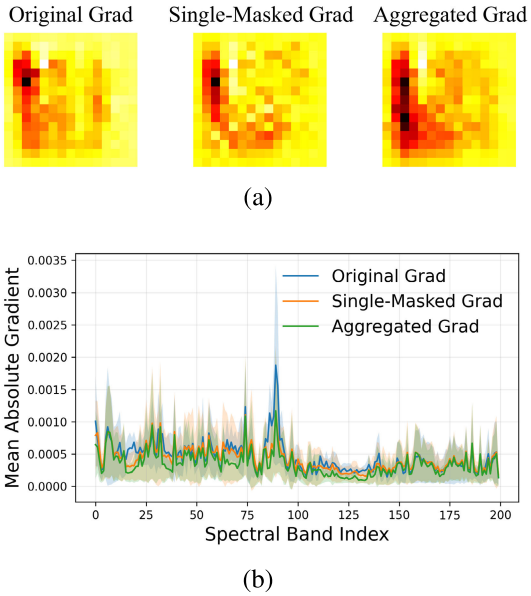


Fig. 9. (a) Spatial gradient maps at the first attack iteration for three constructions: Original Grad, Single-Masked Grad, and Aggregated Grad. (b) Spectral comparison: per-band mean absolute gradient energy with ± 1 standard-deviation bands, averaged over a representative set of test samples.

2) Gradient Distribution Analysis: To better explain the ablation results, we analyze gradient distributions before and after aggregation. Fig. 9(a) shows spatial gradient maps at the first iteration for three cases: Original Grad (no masking), Single-Masked Grad (one random mask), and Aggregated Grad (average over N random masks). Fig. 9(b) shows the mean absolute gradient energy across spectral bands with ± 1

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TABLE VII

ABLATION STUDY OF THE PROPOSED ATTACK ON INDIAN PINES, SALINAS, AND HOUSTONU 2018 DATASETS. A TICK (✓) MEANS THE CORRESPONDING MODULE IS ENABLED. "*" INDICATES A WHITE-BOX ATTACK SCENARIO, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Constrain condition			Indian Pines (%)					
MS-SGS	RM&GA	SGS	GSCViT	SSFTT	CMT	spectralFormer	HiT	morphFormer
✓			87.27*	24.64	37.91	29.09	31.54	21.75
	✓		93.89*	27.54	43.56	28.85	30.75	21.19
		✓	88.25*	20.69	42.59	24.77	24.34	16.41
✓	✓		94.75*	36.54	47.30	36.48	40.16	31.10
✓		✓	87.66*	25.70	40.46	30.91	33.99	23.58
	✓	✓	94.06*	29.31	44.32	32.18	33.97	23.59
✓	✓	✓	**94.87***	**37.44**	**50.07**	**37.61**	**41.73**	**32.85**
Constrain condition			Salinas (%)					
MS-SGS	RM&GA	SGS	GSCViT	SSFTT	CMT	spectralFormer	HiT	morphFormer
✓			83.14*	49.82	52.36	54.88	48.94	41.66
	✓		**96.13***	52.55	50.34	53.15	47.42	43.82
		✓	94.41*	48.64	42.51	56.86	39.91	34.03
✓	✓		95.86*	60.30	63.81	61.32	58.07	53.49
✓		✓	94.21*	53.45	54.75	61.74	48.56	43.65
	✓	✓	95.70*	53.88	52.19	54.54	48.79	44.85
✓	✓	✓	95.56*	**61.12**	**64.52**	**61.85**	**58.96**	**53.99**
Constrain condition			HoustonU 2018 (%)					
MS-SGS	RM&GA	SGS	GSCViT	SSFTT	CMT	spectralFormer	HiT	morphFormer
✓			91.40*	70.48	68.35	63.55	66.03	67.83
	✓		93.23*	74.15	70.85	63.45	66.74	68.31
		✓	91.52*	72.10	69.51	62.44	65.49	67.91
✓	✓		92.80*	74.25	71.41	64.19	69.27	69.16
✓		✓	91.67*	71.43	69.07	63.52	66.12	68.10
	✓	✓	92.91*	74.82	71.10	65.46	68.59	68.64
✓	✓	✓	**93.27***	**74.88**	**71.91**	**66.93**	**69.61**	**69.32**

standard-deviation ranges. Clear differences can be observed. Original Grad shows concentrated high-magnitude regions. Single-Masked Grad becomes spatially fragmented, with many isolated high-intensity pixels, which indicates strong mask-dependent noise. Aggregated Grad restores spatial coherence and exhibits smoother transitions. In the spectral domain, Single-Masked Grad has the largest variance, with a mean per-band variance of 1.474×10^{-7}. After aggregation, the variance drops to 1.038×10^{-7}. The variance of Aggregated Grad is also lower than that of Original Grad (1.127×10^{-7}). Meanwhile, all three gradients show similar spectral energy trends, but Aggregated Grad has the narrowest variation range. These results show that RM&GA mainly acts as a variance-reduction process. Although a single random mask introduces noise, averaging gradients from multiple masks suppresses mask-specific fluctuations and keeps stable gradient components. This leads to a more reliable update direction that is less dependent on the source model. After aggregation, SGS and MS-SGS further shape the gradients. SGS enforces spatial continuity, while MS-SGS enforces smoothness across neighboring spectral bands and across scales. They remove residual high-frequency artifacts and keep consistent spatial–spectral structures. Together, RM&GA provides a stable gradient direction, and SGS and MS-SGS impose spatial and spectral structure. This explains why the full SSJS achieves the best transferability in Table VII: it may slightly reduce white-box strength, but it produces perturbations that generalize better across different architectures.

E. Hyperparameter Analysis of Attack Performance

We study the effect of key hyperparameters using GSCViT as the source model and SSFTT as the target model, with a perturbation budget of 0.03.

1) Spatial Gaussian Kernel Size: The impact of the size of the Gaussian kernel k on the attack performance is shown in Fig. 10. In the white-box case, increasing k from 3 to 9 reduces the success rate on all datasets, because heavy smoothing removes high-frequency details that GSCViT exploits. In contrast, black-box performance rises steadily (e.g., from 74.88% to 75.56% on HoustonU 2018), suggesting that a wider kernel spreads the perturbation and aids transfer. Balancing these trends and the extra runtime of larger kernels, we set k to 3 as the final choice.

2) Fusion Weight: The impact of the spatial–spectral fusion weight β on attack performance is shown in Fig. 11. In the white-box setting, increasing β from 0.1 to 0.7 leads to a slight improvement on Indian Pines and Salinas, with performance peaking around the mid-range and then stabilizing. On HoustonU 2018, the success rate also exhibits a mild upward trend before eventually plateauing. In contrast, in the black-box setting, the success rates steadily decline as β increases across all three datasets. These results indicate that a larger spatial contribution benefits white-box performance but weakens transferability, so we set β to 0.3 as a tradeoff throughout the experiments.

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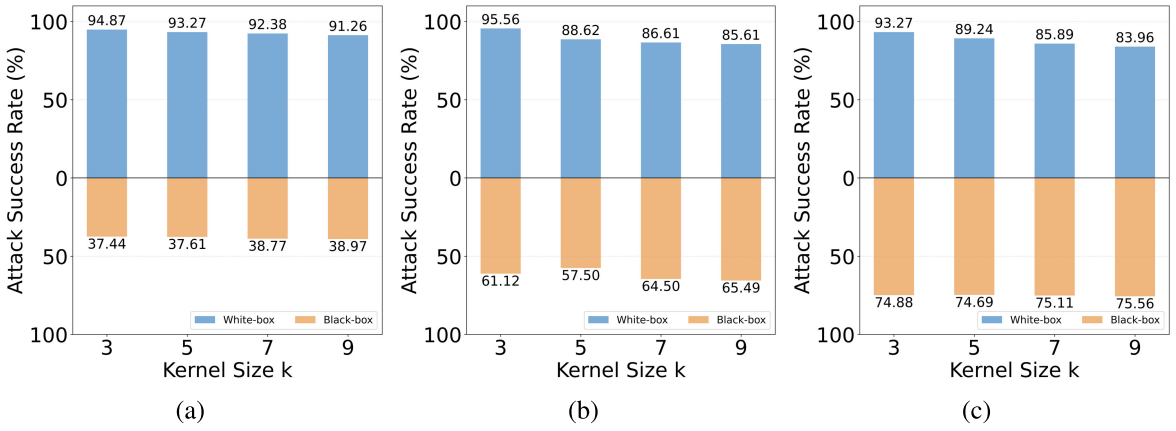


Fig. 10. Impact of the Gaussian kernel size k on the attack success rate for three HSI datasets. (a) Indian Pines. (b) Salinas. (c) HoustonU 2018.

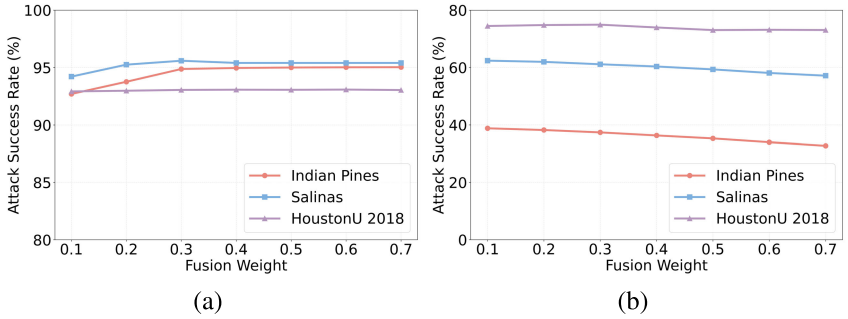


Fig. 11. Impact of the spatial–spectral fusion weight on the attack success rate for three HSI datasets. (a) White-box: adversarial examples tested on GSCViT. (b) Black-box: same examples transferred to SSFTT.

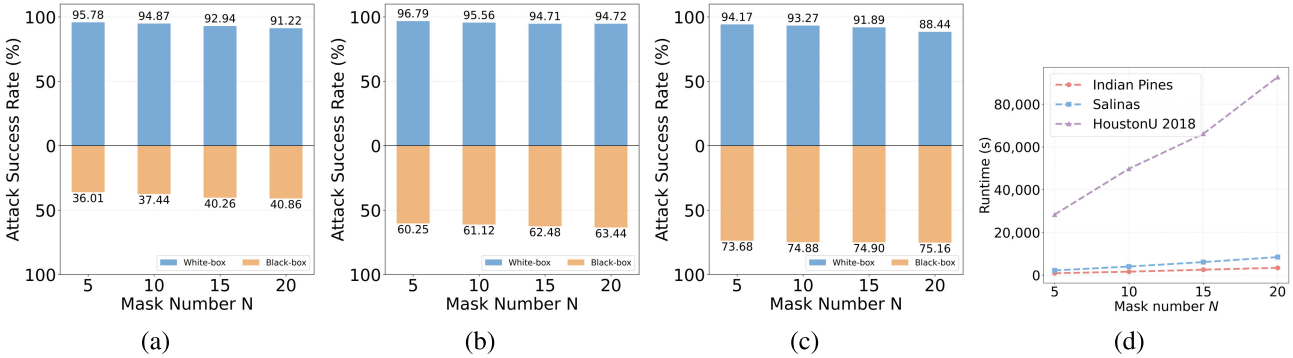


Fig. 12. Impact of the mask number N on the attack success rate and computational cost for three HSI datasets. (a) Indian Pines. (b) Salinas. (c) HoustonU 2018. (d) Average generation runtime (s) for the same adversarial examples.

3) Mask Number: The impact of the mask number N on attack performance and computational cost is shown in Fig. 12. In the white-box scenario, increasing N gradually reduces the effectiveness of the attack across all datasets, although the drop is relatively mild. In contrast, black-box attack success rates show moderate improvements as N increases. However, Fig. 12(d) demonstrates that runtime grows sharply with larger N on a large dataset such as HoustonU 2018. Considering this balance between transferability and computational overhead, setting the mask number to 10 is a good choice.

In conclusion, the results of our hyperparameter study show that a small Gaussian kernel ($k = 3$), a moderate spatial–spectral fusion weight ($\beta = 0.3$), and a mask number of $N = 10$ together provide a favorable tradeoff between white-box accuracy, cross-model transferability, and computational overhead.

F. Visual and Spectral Analysis of Adversarial Examples

We visualize false color adversarial images and plot spectral curves to study the visual quality and spectral behavior of different attack methods. Fig. 13 shows adversarial examples on the Indian Pines, Salinas, and HoustonU 2018 datasets. All methods generate adversarial images that look almost the same as the clean images. This shows that the perturbations are visually hard to notice in the spatial domain. This result agrees with Table VIII, where all methods obtain high PSNR

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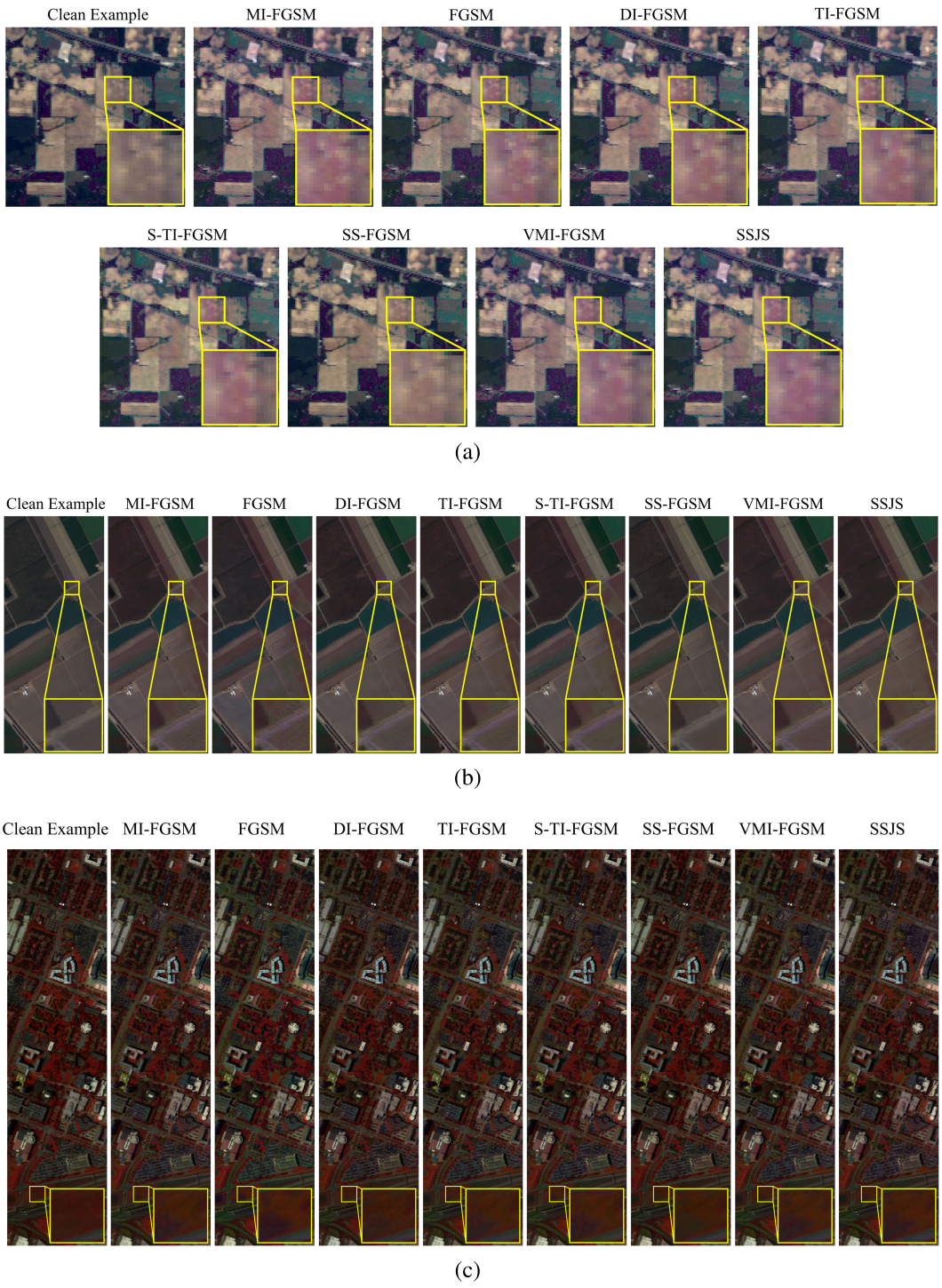


Fig. 13. Visualization of the generated adversarial examples. (a) Indian Pines. (b) Salinas. (c) HoustonU 2018.

and SSIM values using GSCViT as the source model. These results indicate that the perturbations cause only small pixel-level and structural changes.

Visual similarity is not enough for HSIs because spectral signatures must also be preserved. Fig. 14 shows spectral curves of a representative pixel before and after attack. Many baseline methods produce sharp and irregular changes in narrow spectral bands. This reflects strong high-frequency spectral distortion. SSJS produces smoother spectral changes across neighboring bands. The perturbations form continuous cross-band patterns. This behavior is also reflected by the spectral angle mapper (SAM) results in Table VIII. SSJS obtains smaller SAM values than baseline methods. Most SAM values of SSJS fall between 0.03 and 0.09 radians, which indicates better preservation of spectral angles. These results suggest that adversarial examples succeed by creating robust adversarial features rather than by severely destroying spectral integrity.

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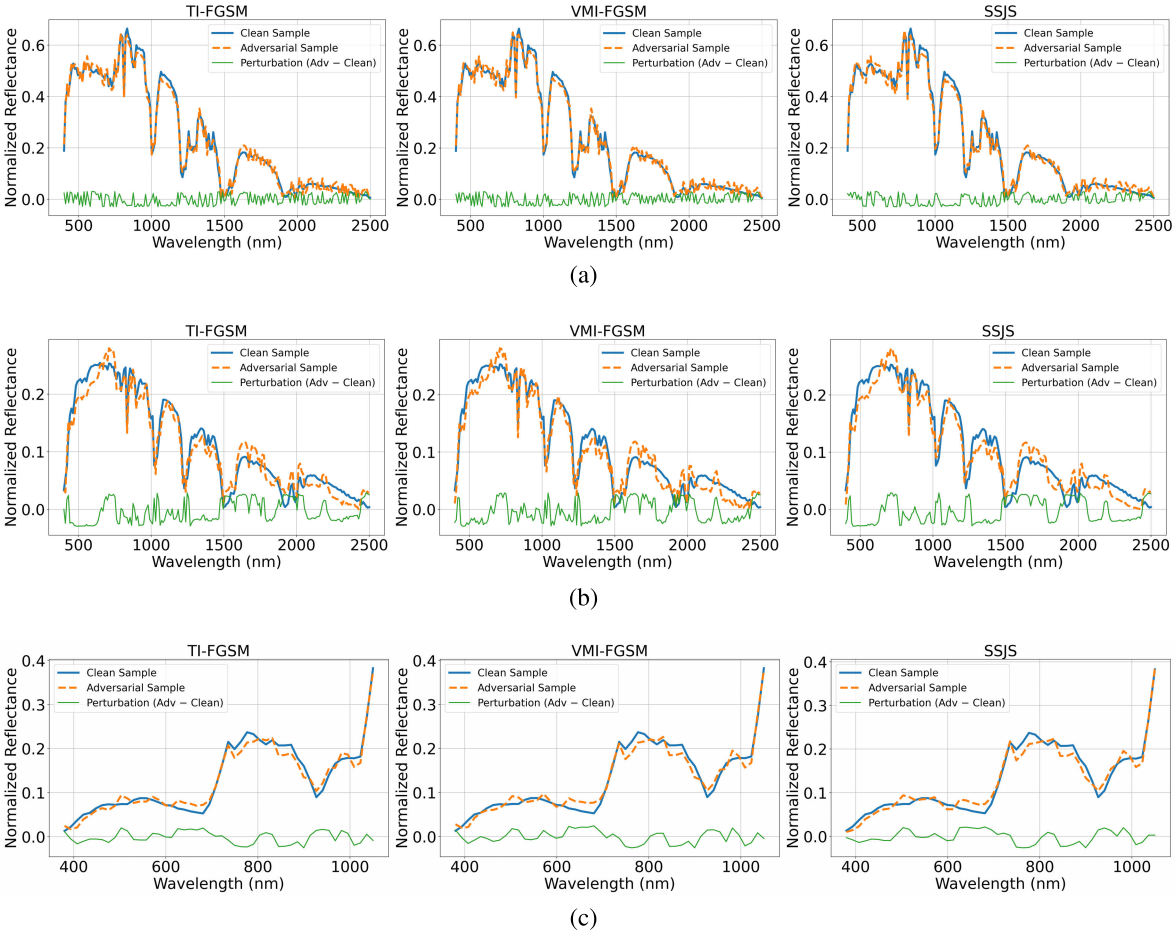


Fig. 14. Spectral profiles of clean, adversarial, and perturbation curves under three attacks for three HSI datasets. (a) Indian Pines. (b) Salinas. (c) HoustonU 2018. Perturbations from baseline methods are high-frequency and spiky, whereas the perturbation from our SSJS method is visibly smoother and more continuous. This suggests that SSJS modifies the spectrum in a more structured manner, thereby enhancing the transferability of the attack to unknown black-box models.

TABLE VIII
PERCEPTUAL AND SPECTRAL QUALITY OF ADVERSARIAL EXAMPLES GENERATED BY DIFFERENT ATTACKS

Dataset	Method	PSNR (dB) ↑	SSIM ↑	SAM (rad) ↓
Indian Pines	VMI-FGSM	32.86	0.9009	**0.0310**
	TI-FGSM	32.85	0.9036	0.0311
	SSJS	**32.93**	**0.9254**	0.0312
Salinas	VMI-FGSM	32.76	0.8925	0.0690
	TI-FGSM	31.05	0.8826	0.0691
	SSJS	**32.84**	**0.9430**	**0.0689**
HoustonU 2018	VMI-FGSM	32.50	0.9168	0.0899
	TI-FGSM	32.51	0.9180	0.0890
	SSJS	**32.68**	**0.9250**	**0.0889**

V. CONCLUSION

In this article, we propose a transferable adversarial attack method named SSJS for HSI classification. SSJS simultaneously enhances perturbation smoothness along spectral and spatial dimensions via multiscale spectral Gaussian smoothing and spatial Gaussian smoothing. Experiments show that SSJS outperforms state-of-the-art methods and increases the average transfer attack success rate by 5.46%, 6.55%, and 2.48% on three benchmark datasets. Crucially, adversarial examples crafted by SSJS also exhibited strong resilience to preprocessing defenses, confirming their potency and advanced nature. By revealing these vulnerabilities with unprecedented clarity, our research underscores the urgent need for more robust HSI classification models. Future work may study applying this attack framework to adversarial training and to physical-world scenarios. Another direction is to design an adaptive spectral smoothing scheme that links kernel parameters to sensor spectral resolution such as FWHM. This wavelength-aware design may improve generalization across sensors with different spectral sampling intervals and reduce reliance on empirical parameter selection.

REFERENCES

[1] A. Plaza et al., "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Jan. 2009.
[2] W. Xu, M. J. Wooster, and C. S. B. Grimmond, "Modelling of urban sensible heat flux at multiple spatial scales: A demonstration using airborne hyperspectral imagery of Shanghai and a temperature–emissivity separation approach," *Remote Sens. Environ.*, vol. 112, no. 9, pp. 3493–3510, Sep. 2008.
[3] D. A. Roberts, D. A. Quattrochi, G. C. Hulley, S. J. Hook, and R. O. Green, "Synergies between VSWIR and TIR data for the urban environment: An evaluation of the potential for the hyperspectral infrared imager (HyspIRI) decadal survey mission," *Remote Sens. Environ.*, vol. 117, pp. 83–101, Feb. 2012.

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[4] B. Lu, P. Dao, J. Liu, Y. He, and J. Shang, "Recent advances of hyperspectral imaging technology and applications in agriculture," *Remote Sens.*, vol. 12, no. 16, p. 2659, Aug. 2020.
[5] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.
[6] A. Mohan and M. Venkatesan, "HybridCNN based hyperspectral image classification using multiscale spatiospectral features," *Infr. Phys. Technol.*, vol. 108, Aug. 2020, Art. no. 103326.
[7] Y. Ge, X. Zhang, P. M. Atkinson, A. Stein, and L. Li, "Geoscience-aware deep learning: A new paradigm for remote sensing," *Sci. Remote Sens.*, vol. 5, Jun. 2022, Art. no. 100047.
[8] Y. Qin, N. Carlini, G. W. Cottrell, I. Goodfellow, and C. Raffel, "Imperceptible, robust, and targeted adversarial examples for automatic speech recognition," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 5231–5240.
[9] A. Ilyas, S. Santurkar, D. Tsipras, L. Engstrom, B. Tran, and A. Mądry, "Adversarial examples are not bugs, they are features," in *Proc. Adv. neural Inf. Process. systems(NeurIPS)*, vol. 32, 2019, pp. 125–136.
[10] X. Yuan, P. He, Q. Zhu, and X. Li, "Adversarial examples: Attacks and defenses for deep learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2805–2824, Sep. 2019.
[11] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.
[12] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, Jul. 2018, pp. 2137–2146.
[13] Y. Liu, X. Chen, C. Liu, and D. Song, "Delving into transferable adversarial examples and black-box attacks," 2016, *arXiv:1611.02770*.
[14] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.
[15] F. Tramèr, A. Kurakin, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, "Ensemble adversarial training: Attacks and defenses," 2017, *arXiv:1705.07204*.
[16] A. Mądry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. 6th Int. Conf. Learn. Represent. (ICLR)*, May 2017, pp. 1–8.
[17] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.
[18] C. Liu et al., "Boosting adversarial transferability for hyperspectral image classification using 3D structure-invariant transformation and weighted intermediate feature divergence," 2025, *arXiv:2506.10459*.
[19] C. Shi, M. Zhang, Z. Lv, Q. Miao, and C.-M. Pun, "Universal object-level adversarial attack in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532714.
[20] Y. Dong, T. Pang, H. Su, and J. Zhu, "Evading defenses to transferable adversarial examples by translation-invariant attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4312–4321.
[21] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.
[22] H. Zhang, Y. Li, Y. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery using a dual-channel convolutional neural network," *Remote Sens. Lett.*, vol. 8, no. 5, pp. 438–447, May 2017.
[23] J. Ma, Y. Zou, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Spatial pooling transformer network and noise-tolerant learning for noisy hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509719.
[24] L. Sun, H. Zhang, Y. Zheng, Z. Wu, Z. Ye, and H. Zhao, "MASSFormer: Memory-augmented spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516415.
[25] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.
[26] H. Shi, Y. Zhang, G. Cao, and D. Yang, "MHCFormer: Multiscale hierarchical conv-aided Fourierformer for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.
[27] Y. Dong et al., "Boosting adversarial attacks with momentum," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9185–9193.
[28] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial machine learning at scale," 2016, *arXiv:1611.01236*.
[29] Y. Li, L. Li, L. Wang, T. Zhang, and B. Gong, "NATTACK: Learning the distributions of adversarial examples for an improved black-box attack on deep neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 3866–3876.
[30] W. Brendel, J. Rauber, and M. Bethge, "Decision-based adversarial attacks: Reliable attacks against black-box machine learning models," 2017, *arXiv:1712.04248*.
[31] W. Chen, Z. Zhang, X. Hu, and B. Wu, "Boosting decision-based black-box adversarial attacks with random sign flip," in *Proc. Eur. Conf. Comput. Vis.*, vol. 12360, 2020, pp. 276–293.
[32] L. Chen et al., "Attack selectivity of adversarial examples in remote sensing image scene classification," *IEEE Access*, vol. 8, pp. 137477–137489, 2020.
[33] H. Li et al., "Adversarial examples for CNN-based SAR image classification: An experience study," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1333–1347, 2021.
[34] C. Du, C. Huo, L. Zhang, B. Chen, and Y. Yuan, "Fast C&W: A fast adversarial attack algorithm to fool SAR target recognition with deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[35] C. Shi et al., "Multifeature collaborative adversarial attack in multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5631815.
[36] L. Liu, Z. Xu, D. He, D. Yang, and H. Guo, "Local pixel attack based on sensitive pixel location for remote sensing images," *Electronics*, vol. 12, no. 9, p. 1987, Apr. 2023.
[37] C. Shi, Y. Liu, M. Zhao, C.-M. Pun, and Q. Miao, "Attack-invariant attention feature for adversarial defense in hyperspectral image classification," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109955.
[38] Y. Xu, B. Du, and L. Zhang, "Self-attention context network: Addressing the threat of adversarial attacks for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 8671–8685, 2021.
[39] Y. Xu, Y. Xu, H. Jiao, Z. Gao, and L. Zhang, "S^3ANet: Spatial–spectral self-attention learning network for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512913.
[40] Z. Wang, H. Guo, Z. Zhang, W. Liu, Z. Qin, and K. Ren, "Feature importance-aware transferable adversarial attacks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7639–7648.
[41] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[42] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.
[43] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.
[44] S. Jia, Y. Wang, S. Jiang, and R. He, "A center-masked transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510416.
[45] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.
[46] X. Wang and K. He, "Enhancing the transferability of adversarial attacks through variance tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1924–1933.
[47] C. Xie et al., "Improving transferability of adversarial examples with input diversity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2730–2739.
[48] C. Guo, M. Rana, M. Cisse, and L. Van Der Maaten, "Countering adversarial images using input transformations," 2017, *arXiv:1711.00117*.
[49] W. Xu, D. Evans, and Y. Qi, "Feature squeezing: Detecting adversarial examples in deep neural networks," 2017, *arXiv:1704.01155*.



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